

# Joseph Fabricio Vergel Becerra

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*Over 7+ years of experience designing and building scalable data architectures, distributed pipelines, and analytics systems for AI and enterprise automation. Specialized in Data Engineering, Machine Learning Infrastructure, and Cloud Data Pipelines (GCP). I have led data science and backend development teams at guane enterprises, solving problems for various industrial sectors including logistics, energy, and business process outsourcing.*

## PROFESSIONAL EXPERIENCE

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### **guane emerging technologies**

**Medellín, Colombia**

#### **Lead Machine Learning Engineer | Lead AI Automation Engineer**

*January 2018 – Present*

- Migrated traditional Natural Language Processing (NLP) pipelines to agentic workflows (Generative AI) using PostgreSQL, FastAPI, Celery, RabbitMQ, and Redis, to automate email-based quotes for a Third-Party Logistics (3PL) broker, processing 10k+ daily requests.
- Developed RESTful API web services under an event-driven architecture ("asynchronous task queuing") for the identification, segmentation, extraction, normalization, and standardization of information from scanned and/or digital documents in the logistics and business process outsourcing (BPO) sectors.
- Built a distributed processing pipeline with Python and RAY for calculating regulated electric service quality indicators (SAIFI, SAIDI, among others), designed by the Energy and Gas Regulation Commission (CREG) of Colombia, for the evaluation and auditing of Local Distribution Systems (LDS).
- Built distributed data pipelines with Google Cloud Dataflow and Apache Beam for real-time analytics and sales reporting integrated with Salesforce CRM, reducing manual reporting time by 80%.
- Built Techindicators, a Python library for generating composite indicators for the technical analysis of any financial asset (securities, commodities, currencies, cryptocurrencies) to support decision-making in the precious metals and mining market.

**Medellín, Colombia**

#### **Data Product Manager**

*January 2022 – October 2023*

- Contributed to and guided the design, planning, development, and agile (Scrum) management of the following projects:
  - Generative AI-based evaluation platform for Australian visa applications (classes 485 and 500).
  - Intelligent Document Management System (DMS) for the civil works auditing sector.
  - Collaborative email platform for the logistics sector.
  - Cloud-based forecasting suite for predicting electricity generation and demand.

### **Panamerican Business School**

**Guatemala**

#### **Instructor - Master's in Big Data and Business Analytics**

*January 2023 – June 2025*

- Taught the following courses:
  - "Introduction to Python" and "Python Business Applications"
  - "Introduction to the R Programming Language" and "R's Applications"
  - "Machine Learning"

### **Universidad de Antioquia**

**Medellín, Colombia**

#### **Instructor - Data Science**

*October 2019 – Jan. 2020*

- Profesor de ciencia de datos para el curso introductorio de inteligencia artificial en el ámbito médico "HealthCare - 4.0".

## OTHER FEATURED PROJECTS

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- Implemented a data flow for the training, deployment, inference and register of machine learning models with probabilistic uncertainty quantification (conformal prediction) with Scikit-Learn and MAPIE, for base transportation cost for the Over-the-Road (OTR) modality, through a RESTful service (FastAPI) with asynchronous task queues (Celery) and WebSocket communication.
- Customer segmentation using manifold learning and density-based clustering techniques for logistics, public gas and electric utility companies.
- Semi-supervised credit scoring based on manifold learning (Isomap, MDS and Autoencoders), hierarchical-based clustering (HDBScan), parametric wrapper (MLP: Multi-Layer Perceptron) and Multi-factor Scoring, to score unbanked customers based on their utility bill payment histories.
- Training and fine-tuning of deep learning models (BiLSTM and transformers) for Natural Language Processing (NLP) with batch and request-response inference, for logistics sector platforms.

## EDUCATION

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### UNIVERSIDAD DE ANTIOQUIA

**Medellín, Colombia**

#### *Master's in Physics*

*December 2020*

Project: Experimental application of energy-based models (EBM) to deep learning architectures such as transformers and recurrent neural networks (RNN) (Degree pending).

### UNIVERSIDAD INDUSTRIAL DE SANTANDER

**Bucaramanga, Colombia**

#### *Bachelor's in Physics*

*December 2016*

Project: Interactive meshing and curvature analysis of a sparse point cloud in three-dimensional space obtained by the fringe projection technique. Project GPA: 4.0.

## TECHNICAL SKILLS

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### Programming

- Python
- SQL
- Go
- Bash
- R
- Rust

### Data Engineering

- Celery
- Apache Airflow
- Apache Beam
- Pandas
- RAY
- PySpark
- RabittMQ
- FastAPI
- Redis
- PostgreSQL

### Machine Learning

- Scikit-Learn
- Pytorch
- Tensorflow
- MLflow
- Weights & Biases
- Scipy
- Statsmodels

### Infrastructure

- Docker
- docker-compose
- Kubernetes
- Skaffold
- Github actions
- Kubeflow

### Cloud Engineering

- VertexAI
- CloudStorage
- CloudSQL
- Cloud Run
- Cloud Tasks
- Pub/Sub
- Dataflow